⚠️ **TRAINING EXAMPLE — NOT A REAL IEC 62304 DELIVERABLE**

This project is a **training site, not a real medical device** under any regulatory definition. The device described in this document — “SentinelFlow 500” — is entirely **fictional**, invented for this course: nothing here is a genuine IEC 62304 deliverable, a real regulatory submission, or evidence of any real organisation’s compliance with anything. It illustrates the *kind* of document a real SaMD/SiMD project would produce. See the **document register** for the full explanation.

Clause: 5.2 — Software Requirements Analysis · **Register:** Device Example Register

Document ID: SOP-03 · **Device:** SentinelFlow 500 (fictional) · **Safety class:** C
Status: SOP — template (Procedure not yet written for SentinelFlow 500)

> **Fictional worked example.** See SOP-01 for the disclaimer that applies to this whole document set, and the Device Example Register for the SentinelFlow 500 device description and what "template" status means here.

1. Purpose

This SOP defines how SentinelFlow 500's software requirements are derived from system-level requirements, documented, and kept traceable — the Software Requirements Specification (SRS) process for Clause 5.2.

2. Scope

Applies to all functional, non-functional, interface, and security requirements for SentinelFlow 500's control software (Class C — see SOP-01 §7.3). Covers Clause 5.2 in full, including the requirement to include risk control measures in the requirements set and to re-evaluate the device risk analysis.

3. Responsibilities

ROLE	RESPONSIBILITY UNDER THIS SOP
Software Development Lead	Owns the SRS; ensures every requirement is traceable to a system requirement
QA/RA	Reviews the SRS for completeness, consistency, correctness and testability before design begins
Risk Manager	Confirms risk control measures from SOP-11 are represented in the SRS; re-evaluates the device risk analysis against new/changed requirements

4. Definitions & Abbreviations

TERM	MEANING
SRS	Software Requirements Specification
Traceability	The recorded link from a software requirement to its source system requirement, and forward to the design and test that satisfy it
Risk control measure	A requirement whose purpose is to reduce a risk identified in the risk management file

5. References

- IEC 62304:2006+AMD1:2015, §5.2
- **SOP-01 — General Requirements & Classification**
- **SOP-04 — Software Architecture**: consumes this SRS as its input
- **SOP-11 — Software Risk Management File**: source of the risk control measures §5.2.3 requires in the SRS

6. What the standard requires

REF	TITLE	APPLIES AT CLASS C	WHAT MUST BE PRODUCED
5.2.1	Define and document software requirements from system requirements	Yes	Documented software system requirements, derived from the system level requirements
5.2.2	Software requirements content	Yes	Software requirements content: functional and capability, inputs and outputs, interfaces, alarms and messages, security, user interface, data and database, installation and acceptance, operation and maintenance, IT-network aspects, and regulatory requirements
5.2.3	Include risk control measures in software requirements	Yes	Risk control measures implemented in software, included in the requirements
5.2.4	Re-evaluate medical device risk analysis	Yes	Re-evaluated and updated medical device risk analysis
5.2.5	Update requirements	Yes	Re-evaluated and updated requirements, including system requirements
5.2.6	Verify software requirements	Yes	Documented verification that the software requirements implement the system requirements, do not contradict each other, avoid ambiguity, permit test criteria, are uniquely identifiable and are traceable

(Quoted directly from `applicability.json` , identical in content to the self-referential set's 03.)

7. Procedure

Not yet written for SentinelFlow 500 — see the [Device Example Register](#) for what "template" status means. Once complete, this section would give SentinelFlow 500's actual requirements — e.g. the occlusion-alarm response time, the dose-rate validation range, the alarm escalation behaviour — the device-specific answer to each row of the table in §6.

5.2.1–5.2.2 — Requirements derived from system level, and their required content

[Template.]

5.2.3 — Risk control measures included in requirements

[Template — see SOP-11 for the risk control measures this section would incorporate.]

5.2.4–5.2.5 — Re-evaluate and update the medical device risk analysis and requirements

[Template.]

5.2.6 — Verify software requirements

[Template.]

8. Records

RECORD	PRODUCED BY	RETAINED IN
Software Requirements Specification, current version	Software Development Lead	Project technical file
Requirements review record	QA/RA	Project technical file
Traceability matrix (system → software requirements)	Software Development Lead	Project technical file

9. Revision History

Illustrative only — SentinelFlow 500 has no real revision history.

VERSION	STATUS	DESCRIPTION
1.0	Template	Structure and requirements table complete; Procedure section pending